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Department of Artificial Intelligence (AI) and Data Science

"A Decentralized IoT-Blockchain Framework with Customized Smart Contracts for Transparent and Traceable Agricultural Supply Chains"

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Agenda

- Introduction
- Problem Statement
- Objectives
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- Methodology & Flow Chart of Project Work
- Software & Hardware Requirement Specification
- Use Case Diagram
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Introduction

- **Agricultural supply chains** involve multiple stakeholders, making product origin, quality, transactions, and farmer payments difficult to track transparently.
- **IoT sensors and blockchain** can provide real-time data and immutable records, but blockchain alone cannot verify whether the incoming sensor data is trustworthy.
- Our proposed system introduces an **AI Trust Layer** to validate IoT data before recording it on the blockchain, while **customized smart contracts** manage traceability, custody, quality, and payment-related transactions.
- The system aims to achieve **transparent, trustworthy, end-to-end agricultural traceability** while improving **payment transparency and fairness for farmers**.

Problem Statement

Traditional agricultural supply chains rely on fragmented, siloed record-keeping that leaves farmers, processors, distributors, retailers and consumers without real-time visibility or trust in product origin and quality. This lack of transparency enables food fraud, delayed recalls and poor accountability, while existing blockchain-based solutions remain proof-of-concept, still facing scalability, interoperability, smart-contract security and off-chain data-trust challenges. There is a need to design and implement a scalable, secure blockchain system with customized smart contracts that delivers real, end-to-end traceability across the agricultural supply chain.

Objectives

- **AI-Powered IoT Validation:** Filter anomalous, drifted, or spoofed sensor telemetry before on-chain entry using physical plausibility bounds and an **Isolation Forest** ML model.
- **Tamper-Proof Custody Tracking:** Enforce strict forward-only batch lifecycle transitions (REGISTERED → IN_TRANSIT → IN_STORAGE → AT_RETAIL → SOLD) using EVM smart contracts.
- **Automated Cold-Chain SLA Enforcement:** Continuously monitor ambient conditions (temperature, humidity) and automatically compute dynamic penalties for environmental SLA breaches.
- **Fair & Transparent Farmer Payments:** Guarantee secure, intermediary-free escrow settlements with direct-to-farmer payouts in integer currency units (paise).
- **Consumer Verification & Auditability:** Enable instant QR-code provenance inspection for consumers and real-time quarantine audit trails for all supply chain participants.

Scope

- **Farmer-Centric Supply Chain:** Direct market access with fair grower pricing.
- **AI Sensor Validation:** Pre-chain filtering of faulty/spoofed IoT telemetry via ML.
- **Blockchain Traceability:** Immutable batch provenance and cryptographic audit logs.
- **Custom Smart Contracts:** Enforces SLA penalty rules and custody handoffs in Solidity.
- **Product & Custody Tracking:** Strict sequential transitions (REGISTERED to SOLD).
- **Transparent Records:** Tamper-evident ledger of all events and hashes.
- **Farmer Payment Transparency:** Automated escrow releases with zero unfair deductions.
- **QR Farm-to-Consumer:** Instant consumer mobile scan for origin and trust score

Literature Survey

Ref No	Paper Title	Author(s) & Year	Focus	Key Contribution	Limitation
1	**Blockchain Technology for Global Supply Chain Management: A Survey of Applications, Challenges, Opportunities and Implications**	Dudczyk, Dunston & Crosby (2024)	Blockchain in Global Supply Chains	Reviews blockchain applications and benefits for global supply-chain transparency and traceability.	Lacks agriculture-specific evaluation.
2	**Revolutionizing Agri-Food Supply Chain Management with Blockchain-Based Traceability and Navigation Integration**	Aggarwal, Rani, Rani & Sharma (2024)	Blockchain in Agri-food	Provides blockchain-based traceability and navigation for food provenance and consumer trust.	Scalability concerns in large-scale deployment.
3	**High-Efficiency Blockchain-Based Supply Chain Traceability**	Wu, Jiang & Cao (2023)	Blockchain Traceability	Improves efficiency of blockchain-based supply-chain traceability by reducing search latency and storage overhead.	Limited real-world/large-scale validation.
4	**Integrating Blockchain Technology for Enhanced Transparency and Security in Supply Chains**	Chaker & Damak (2024)	Blockchain for Supply Chain	Discusses blockchain integration for supply-chain transparency and security.	Mostly conceptual; limited quantitative evidence.
5	**Demystifying the Impact of Blockchain on Trust in Emerging and Established Relationships: A Case of Organic Food Supply Chains**	Yavaprabhas, Kurnia, Seyedghorban & Samson (2024)	Blockchain in Organic Food	Examines how blockchain can improve trust in organic food supply-chain relationships.	Findings are context-specific to organic food.

Ref No	Paper Title	Author(s) & Year	Focus	Key Contribution	Limitation
1	**Optimizing Supply Chain Management With IoE and AI**	Agrawal, Aronkar, Palav, Badre, Karumuri & Bagale (2025)	IoE & AI in SCM	Explores AI and Internet of Everything for supply-chain optimization and decision-making.	Lacks empirical validation in agricultural contexts.
2	**Zero-Day Exploits Detection with Adaptive WavePCA-Autoencoder (AWPA) Adaptive Hybrid Exploit Detection Network (AHEDNet)**	Mohamed, Al-Saleh, Sharma & Tejani (2025)	ML-Based Security	Uses a hybrid ML approach for proactive zero-day exploit detection.	Requires high-quality, large training datasets.
3	**A Novel Malware Detection Model in the Software Supply Chain Based on LSTM and SVMs**	Zhou, Li, Fu & Jiao (2024)	ML Malware Detection	Uses LSTM and SVM-based models for malware detection.	Higher model complexity increases deployment cost.
4	**Harnessing Advanced Hybrid Deep Learning Model for Real-Time Detection and Prevention of Man-in-the-Middle Cyber Attacks**	Kandasamy & Roseline (2025)	Deep Learning Security	Applies hybrid deep learning for real-time MITM attack detection and prevention.	Computationally intensive for constrained devices.

Ref No.	Paper Title	Author(s) & Year	Focus	Key Contribution	Limitation
1	**Secure Pharmaceutical Supply Chain Using Blockchain in IoT Cloud Systems**	Mangala, Naveen, Reddy, Buyya, Venugopal, Iyengar & Patnaik (2024)	Blockchain + IoT	Integrates blockchain with IoT-cloud systems for secure authentication and anti-counterfeiting.	High system complexity limits adoption.
2	**Enhancing Supply Chain Resilience and Efficiency Through Internet of Things Integration: Challenges and Opportunities**	Shoomal, Jahanbakht, Componation & Ozay (2024)	IoT-Enabled SCM	Shows how IoT improves supply-chain resilience, efficiency and coordination.	Data privacy risks remain unresolved.
3	**Supply Chain Management Growth With the Adoption of Blockchain Technology (BoT) and Internet of Things (IoT)**	Singh & Dadhich (2023)	Blockchain + IoT	Presents a model showing improved supply-chain coordination and visibility through Blockchain + IoT.	Lacks real implementation and empirical testing.
4	**Blockchain-Driven Security for IoT Networks: State-of-the-Art, Challenges and Future Directions**	Maurya, Rishiwal, Yadav, Shiblee, Yadav, Agarwal & Chaudhry (2025)	IoT Security	Reviews blockchain-based approaches for improving IoT security and data protection.	Introduces computational overhead.
5	**Enhancing IoT Data Security: Using the Blockchain to Boost Data Integrity and Privacy**	Eghmazi, Ataei, Landry & Chevrette (2024)	IoT Data Security	Uses blockchain to improve IoT data integrity, privacy and security.	Scalability with high sensor-data volume is not fully addressed.

Ref No.	Paper Title	Author(s) & Year	Focus	Key Contribution	Limitation
1	**Revolutionizing Agri-Food Supply Chain Management with Blockchain-Based Traceability and Navigation Integration**	Aggarwal, Rani, Rani & Sharma (2024)	Agri-food Traceability	Improves end-to-end food provenance, traceability and consumer trust.	Scalability concerns in large-scale deployment.
2	**Demystifying the Impact of Blockchain on Trust in Emerging and Established Relationships: A Case of Organic Food Supply Chains**	Yavaprabhas, Kurnia, Seyedghorban & Samson (2024)	Organic Food Supply Chain	Demonstrates blockchain's role in strengthening trust within organic food supply chains.	Findings are specific to organic food case studies.
3	**Optimizing Supply Chain Management With IoE and AI**	Agrawal, Aronkar, Palav, Badre, Karumuri & Bagale (2025)	AI in Supply Chain	Explores AI and IoE for supply-chain optimization.	Lacks empirical validation in agricultural contexts.
4	**Enhancing Supply Chain Resilience and Efficiency Through Internet of Things Integration: Challenges and Opportunities**	Shoomal, Jahanbakht, Componation & Ozay (2024)	IoT Supply Chain	Highlights IoT applications for real-time tracking and improved supply-chain resilience, including agri-food chains.	Privacy, interoperability and scalability challenges.
5	**Blockchain Technology for Global Supply Chain Management: A Survey of Applications, Challenges, Opportunities and Implications**	Dudczyk, Dunston & Crosby (2024)	Global Supply Chain	Identifies blockchain's benefits for transparency and traceability across supply chains.	Does not specifically evaluate agricultural applications.

Proposed Methodology

1. **Data Collection:** Collect real-time ambient metrics (temperature, humidity, timestamps) using IoT edge sensors alongside stakeholder batch declarations.

2. **Data Validation:** Deploy the **AI Trust Layer** (physical plausibility checks and an **Isolation Forest** ML model) to detect and quarantine abnormal or spoofed readings.

3. **Blockchain Recording:** Anchor cryptographic audit hashes, trust scores, and verified state transitions on the **blockchain** to guarantee tamper resistance.

4. **Smart Contracts:** Use customized **Solidity smart contracts** to automate cold-chain SLA penalty calculations, custody transfers, and escrow fund management.

5. **Traceability:** Track produce across strict sequential custody stages: **Farmer → Distributor → Retailer →Consumer**.

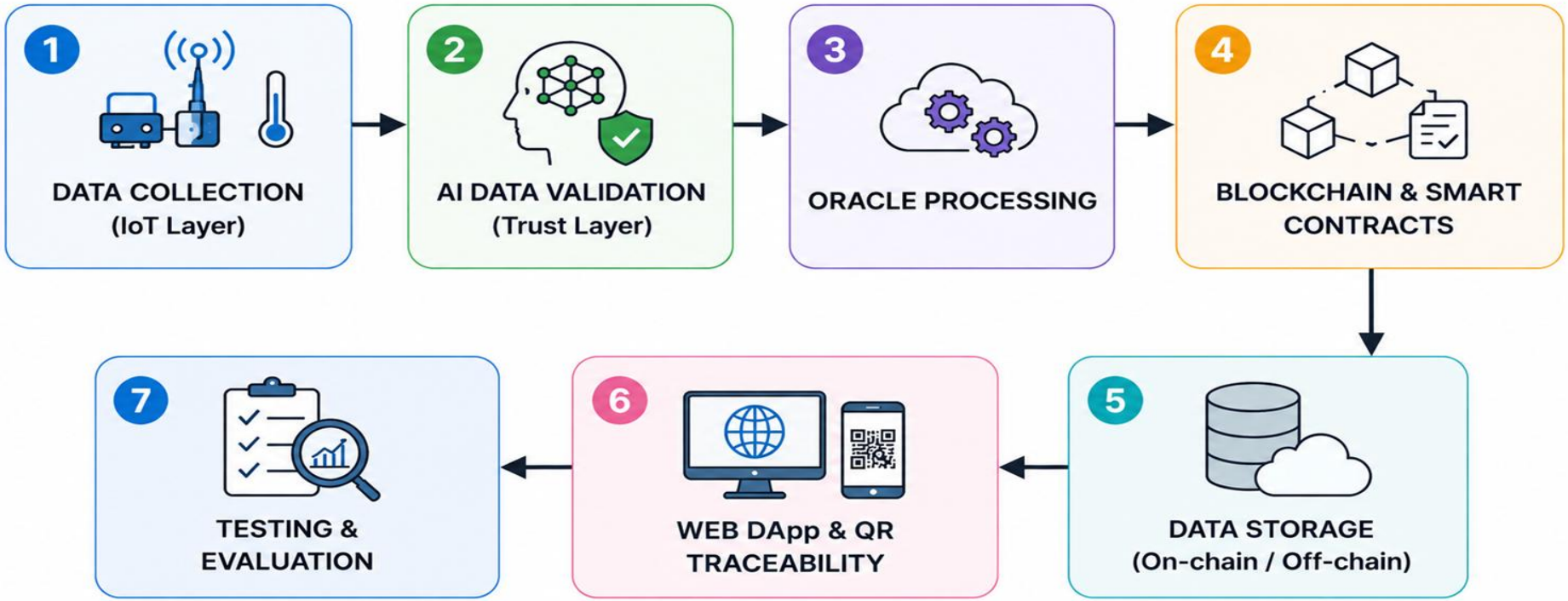
6. **Farmer Payment:** Execute automated, rule-based escrow payouts in integer currency units (**paise**) to guarantee prompt, fair compensation without unfair deductions.

7. **Monitoring & Verification:** Provide stakeholders with live quarantine audit feeds and consumers with instant **QR-code verification** of crop origin and quality trust.

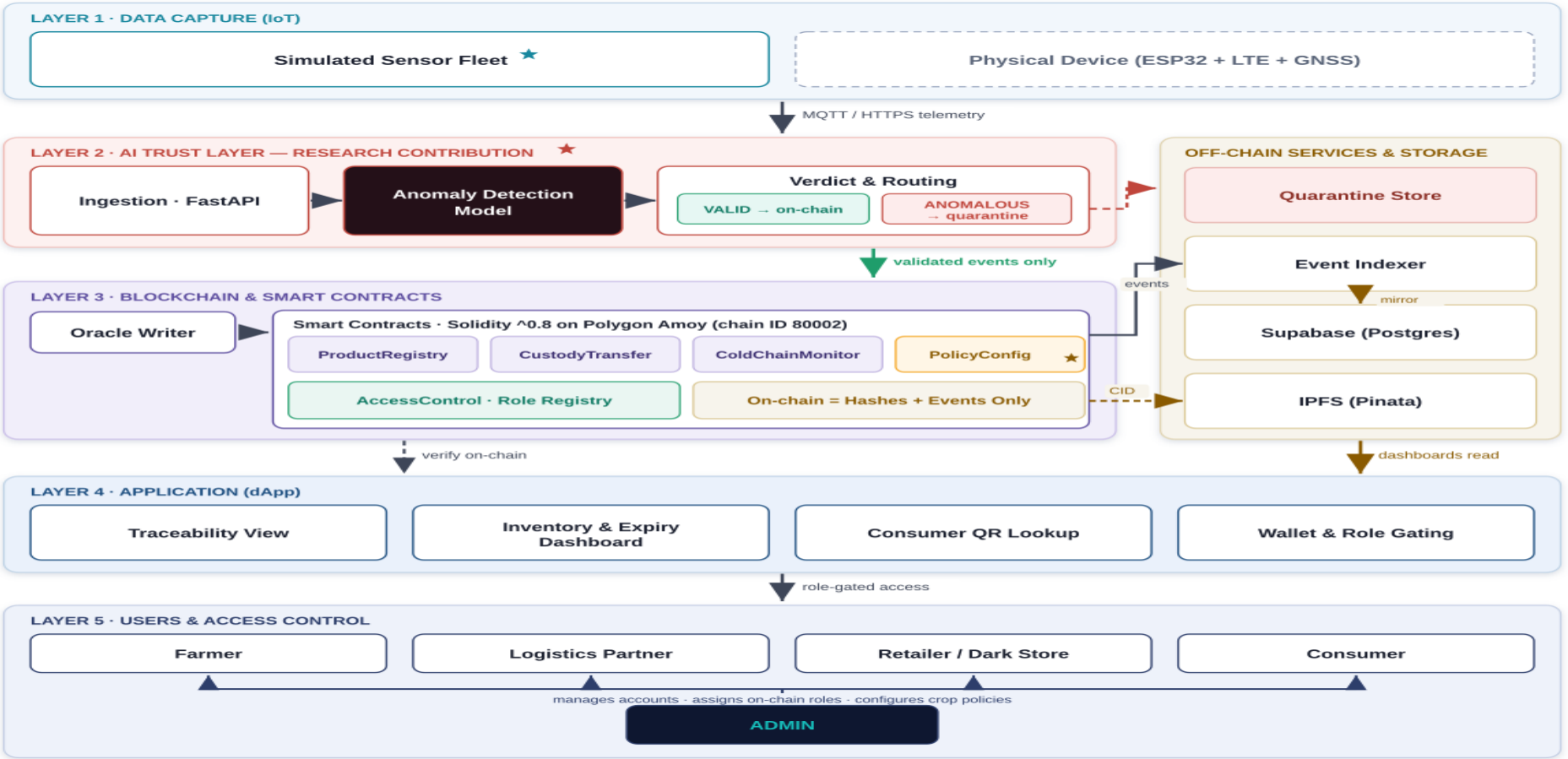
How It Is Better Than Existing Systems

Existing Systems	Our Proposed System (AgriChain)
• Blindly trusts IoT data (Garbage-In, Garbage-Out).	• AI Trust Layer filters and quarantines faulty or spoofed telemetry.
• Centralized records can be altered, delayed, or lost.	• Blockchain immutability guarantees tamper-proof custody history.
• Manual quality checks cause disputes and unfair deductions.	• Smart contracts automatically calculate cold-chain SLA penalties.
• Delayed farmer payments controlled by intermediaries.	• Automated escrow guarantees prompt, direct-to-farmer payouts.
• Consumers have no way to verify product journey or freshness.	• QR-code verification gives consumers full provenance transparency.
• Private logs offer no real-time auditability.	• Live quarantine audit feed provides transparent system oversight.

Flowchart


1
DATA COLLECTION
(IoT Layer)
2
AI DATA VALIDATION
(Trust Layer)
3
ORACLE PROCESSING
4
BLOCKCHAIN & SMART
CONTRACTS
5
DATA STORAGE
(On-chain / Off-chain)
6
WEB DApp & QR
TRACEABILITY
7
TESTING &
EVALUATION

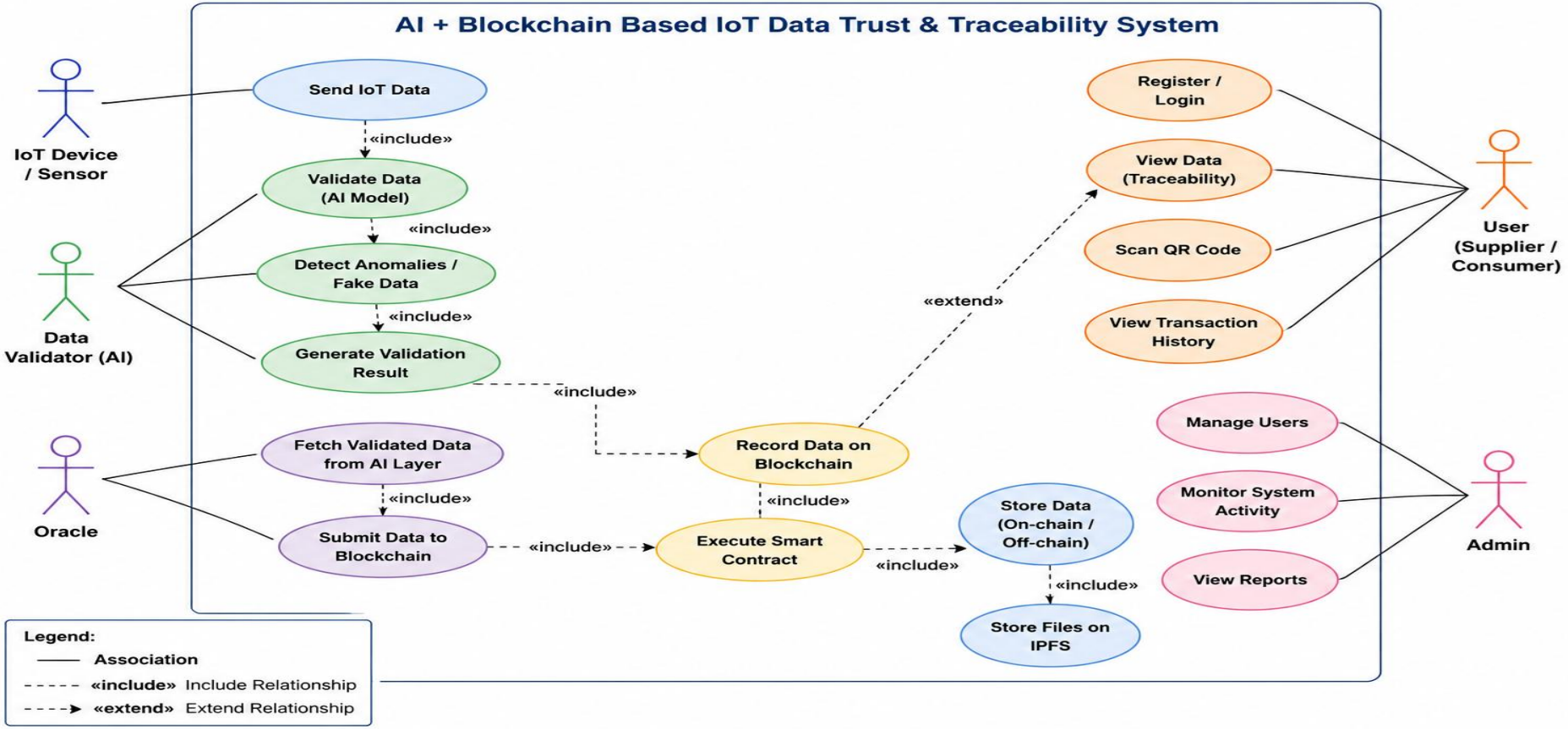
System Architecture


LAYER 1 · DATA CAPTURE (IoT)
Simulated Sensor Fleet
Physical Device (ESP32 + LTE + GNSS)
MQTT / HTTPS telemetry
LAYER 2 · AI TRUST LAYER — RESEARCH CONTRIBUTION
Ingestion · FastAPI
Anomaly Detection Model
Verdict & Routing
VALID → on-chain
ANOMALOUS → quarantine
OFF-CHAIN SERVICES & STORAGE
Quarantine Store
validated events only
Event Indexer
events
mirror
LAYER 3 · BLOCKCHAIN & SMART CONTRACTS
Oracle Writer
Supabase (Postgres)
Smart Contracts · Solidity ^0.8 on Polygon Amoy (chain ID 80002)
ProductRegistry
CustodyTransfer
ColdChainMonitor
PolicyConfig
AccessControl · Role Registry
On-chain = Hashes + Events Only
CID
IPFS (Pinata)
verify on-chain
dashboards read
LAYER 4 · APPLICATION (dApp)
Traceability View
Inventory & Expiry Dashboard
Consumer QR Lookup
Wallet & Role Gating
role-gated access
LAYER 5 · USERS & ACCESS CONTROL
Farmer
Logistics Partner
Retailer / Dark Store
Consumer
manages accounts · assigns on-chain roles · configures crop policies
ADMIN

Software & Hardware Requirements Specification

Category	Component / Specification	Purpose in Architecture
Operating System	Windows 11 / Linux Ubuntu 22.04 LTS	Host environment for local nodes, backend, and development servers
Programming Languages	Python 3.11+, Solidity 0.8.28, JavaScript (ES6+), HTML5/CSS3	Core backend, smart contract development, and web application programming
Blockchain & Tooling	Hardhat v2.22, Polygon Amoy Testnet (Chain ID 80002)	EVM execution environment, compilation, testing, and deployment
Smart Contract Library	OpenZeppelin Contracts v5.0 (AccessControl, Pausable)	Audited role-based access control and emergency circuit breaker patterns
AI & Machine Learning	Scikit-Learn (Isolation Forest), NumPy, SciPy	Unsupervised anomaly detection, multi-dimensional feature engineering
Backend Web Framework	FastAPI, Uvicorn (ASGI), Web3.py v6.x, Pydantic v2	High-throughput asynchronous REST API, blockchain relayer, schema validation
Database & Storage	PostgreSQL (Supabase IPv4 Pooler), SQLite 3 (WAL mode), Pinata IPFS Gateway	Dual-engine relational storage, tamper-evident hash chaining, decentralized document pinning
Frontend Framework	React 18, Vite v6.0, Tailwind CSS v3.4, Lucide React	Responsive multi-role UI dashboards (Farmer, Logistics, Dark Store, Consumer, Admin)
IoT Hardware / Sim	ESP32 DevKit v1, DHT22 Sensor, NEO-6M GPS, Python CLI Simulator	Physical and synthetic multi-fault environmental telemetry streaming

Use Case Diagram


AI + Blockchain Based IoT Data Trust & Traceability System
IoT Device / Sensor
Data Validator (AI)
Oracle
Send IoT Data
«include»
Validate Data (AI Model)
«include»
Detect Anomalies / Fake Data
«include»
Generate Validation Result
«include»
Fetch Validated Data from AI Layer
«include»
Submit Data to Blockchain
«include»
Execute Smart Contract
«include»
Record Data on Blockchain
«include»
Store Data (On-chain / Off-chain)
«include»
Store Files on IPFS
«extend»
Register / Login
View Data (Traceability)
Scan QR Code
View Transaction History
User (Supplier / Consumer)
Manage Users
Monitor System Activity
View Reports
Admin
Legend:
Association
«include» Include Relationship
«extend» Extend Relationship

Key Features

- **IoT-Based Real-Time Monitoring (Simulator) –** Simulates real-time cold-chain environmental streams (temperature, humidity, and location) to monitor perishable crop health continuously during storage and transit.
- **AI Trust Layer –** Serves as an intelligent gatekeeper using physical plausibility rules and an **Isolation Forest** model to detect and quarantine faulty, drifted, or tampered sensor readings before blockchain entry.
- **Blockchain-Based Traceability –** Immutably logs every verified batch lifecycle event and cryptographic audit hash on a decentralized ledger, preventing retrospective record tampering.
- **Customized Smart Contracts –** Governs strict sequential custody transfers (REGISTERED → IN_TRANSIT → IN_STORAGE → AT_RETAIL → SOLD) and automatically calculates penalties for cold-chain SLA breaches.
- **Transparent Farmer Payments –** Locks buyer deposits in automated smart contract escrow, releasing prompt, direct payouts to farmers without intermediary delays or arbitrary quality cuts.
- **End-to-End Supply Chain Visibility –** Provides role-based dashboards with real-time quarantine audit feeds for stakeholders, and instant **QR-code verification** for consumers to check crop origin and freshness.

Design & Implementation

AgriChain

Farm Fresh, Delivered With Trust

Handpicked produce from trusted farmers, delivered fresh to your doorstep.

100% Natural
Pure, healthy and chemical free.

Trusted Farmers
Supporting local farmers and communities.

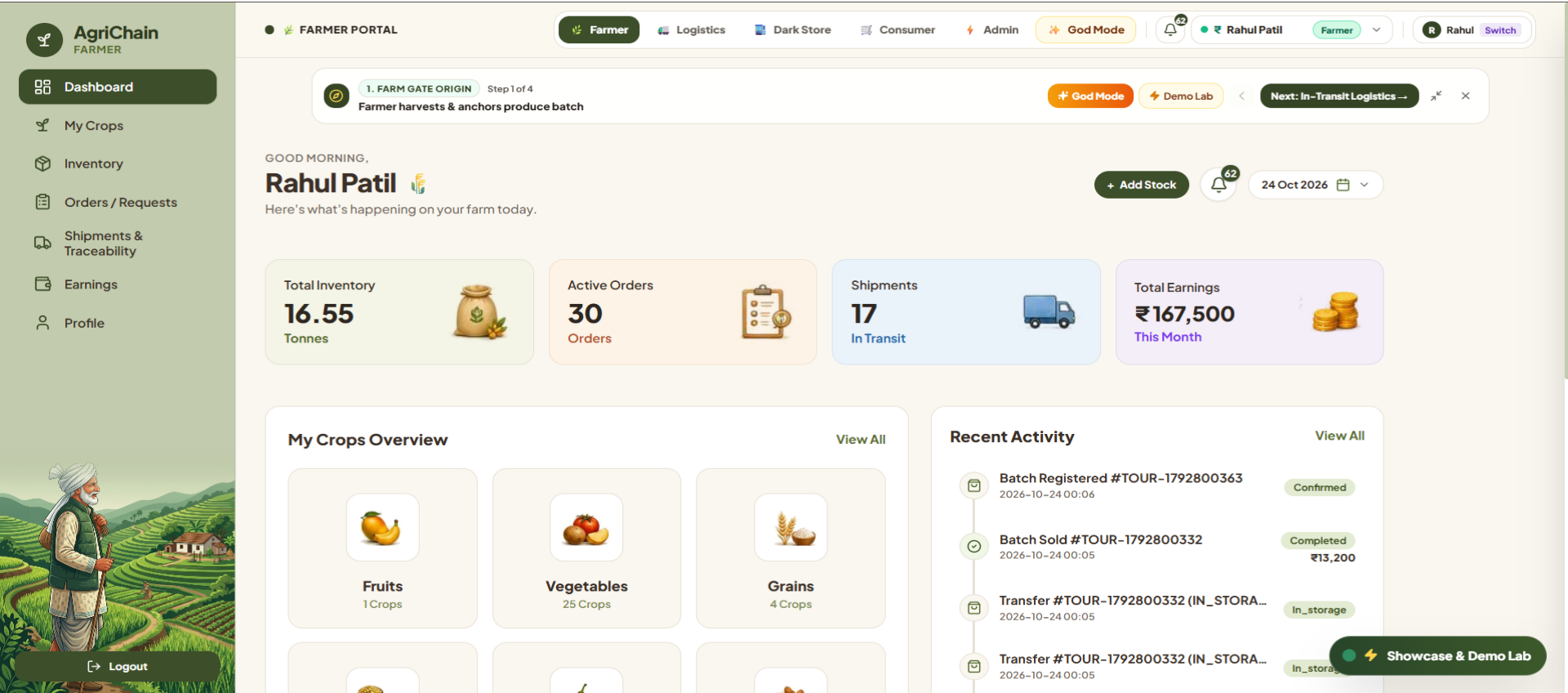
Safe & Secure
Hygienically packed and safely delivered.

Get Started →

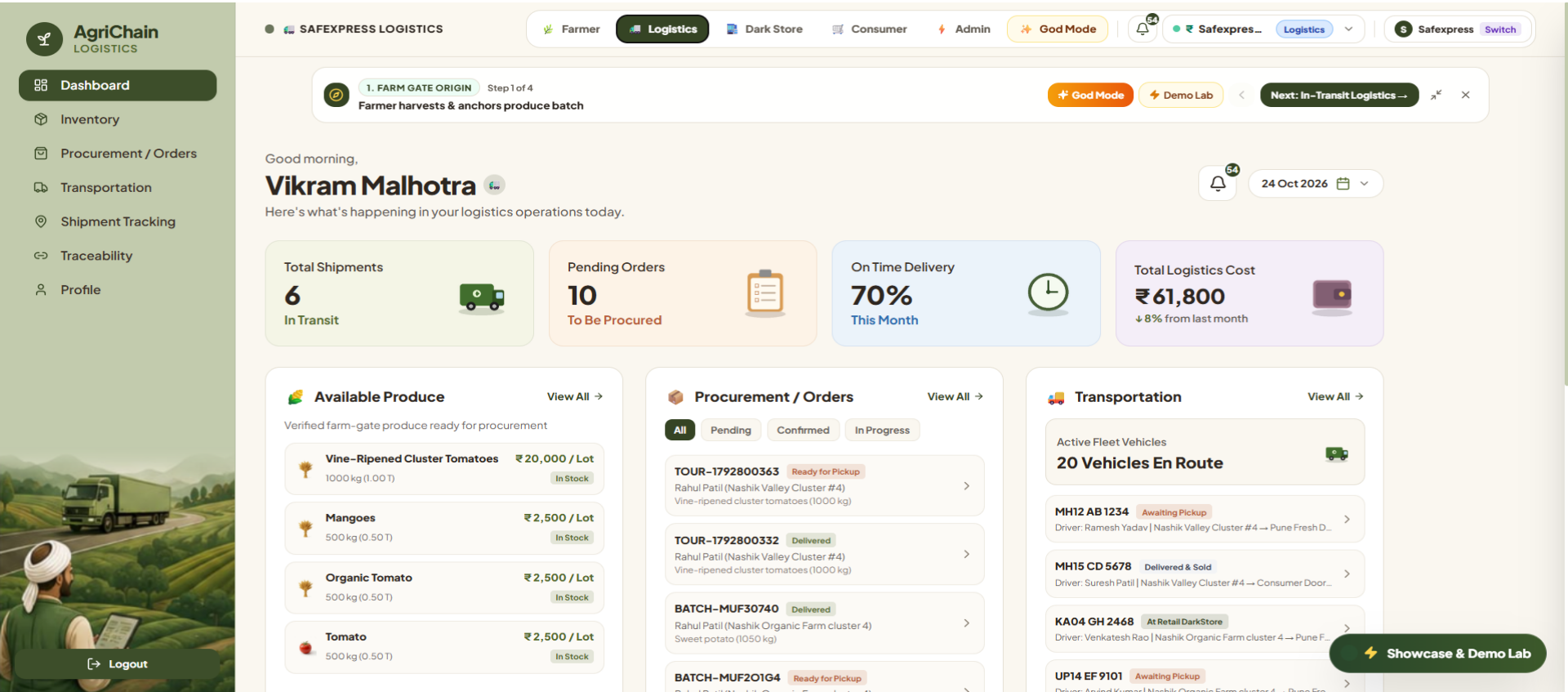
Continue as Guest



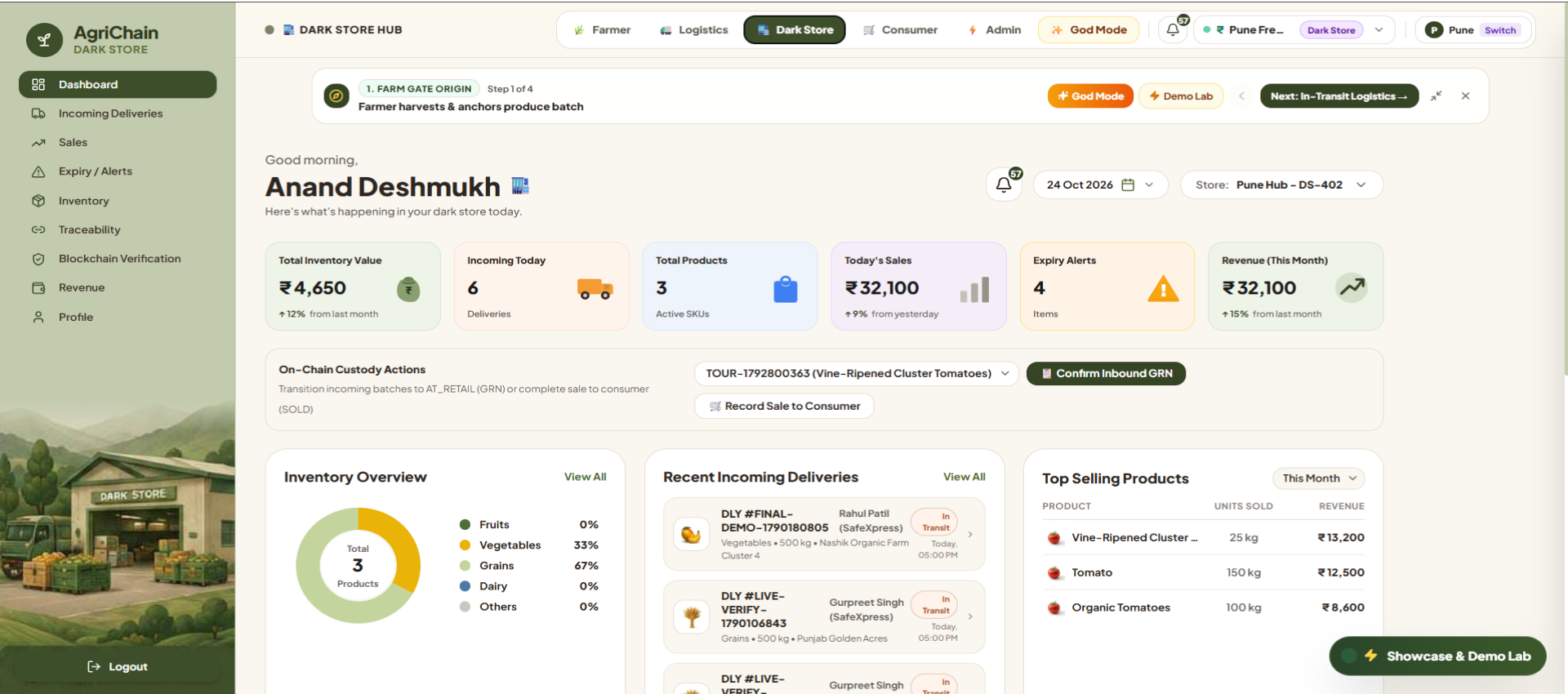
Farmer Dashboard for Agricultural Product and Batch Registration



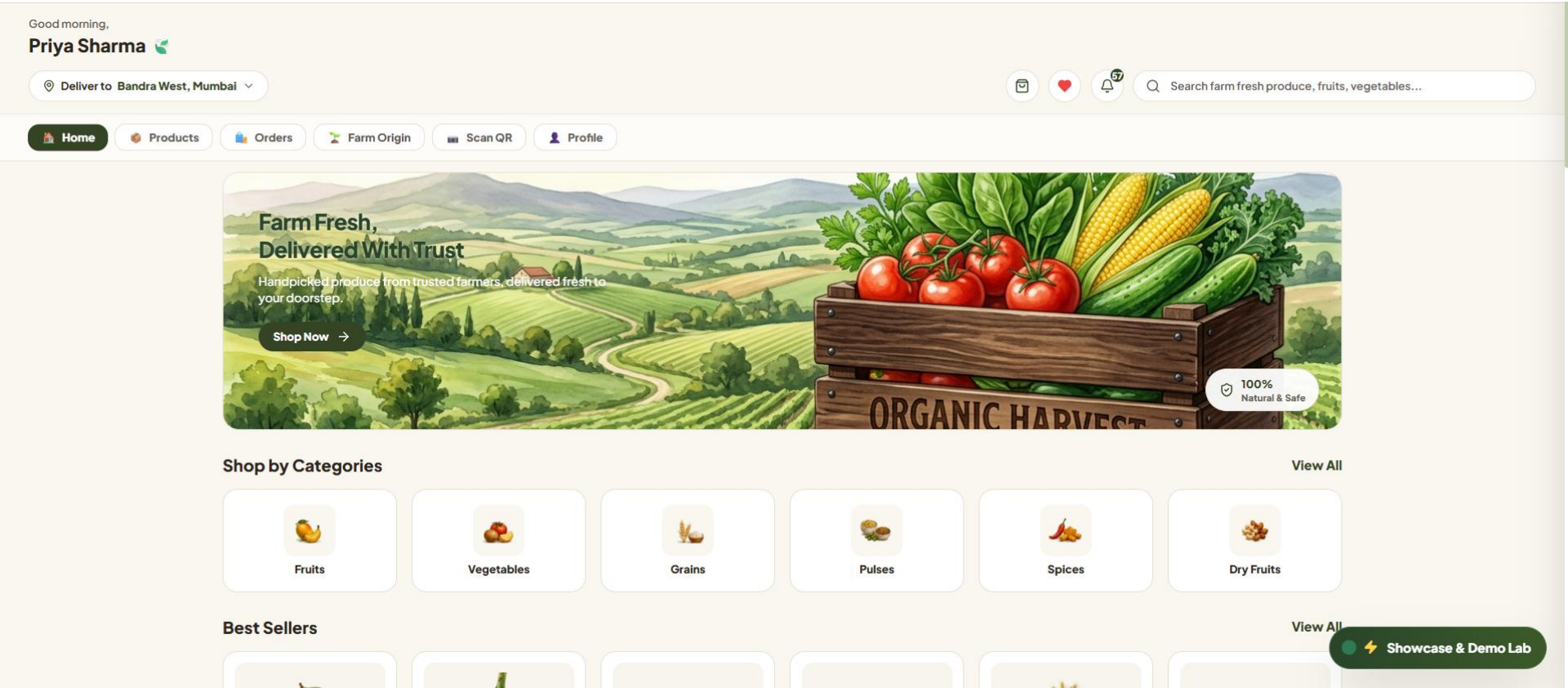
Logistics Dashboard for Product Dispatch and Tracking



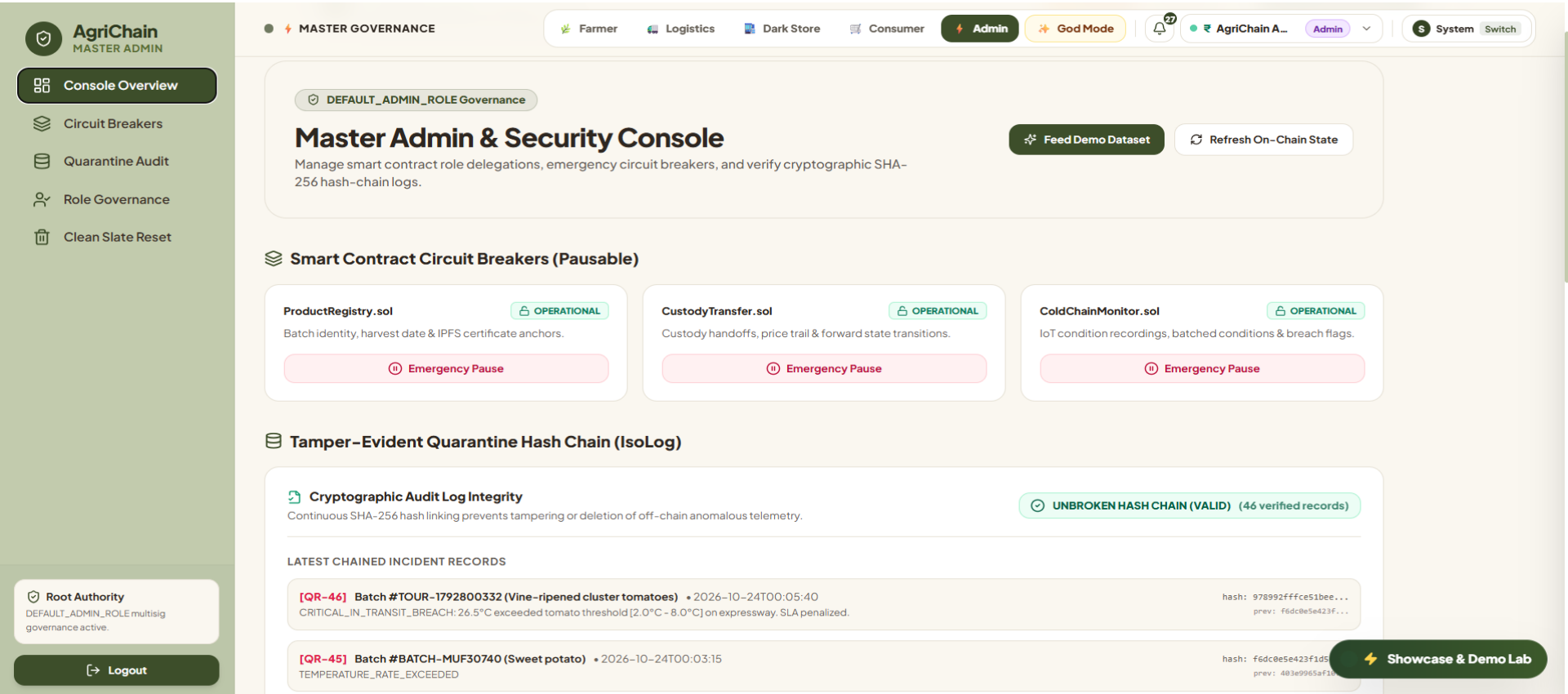
Dark-Store Dashboard for Product Receiving and Inventory Management



Consumer Interface for QR-Based Product Traceability

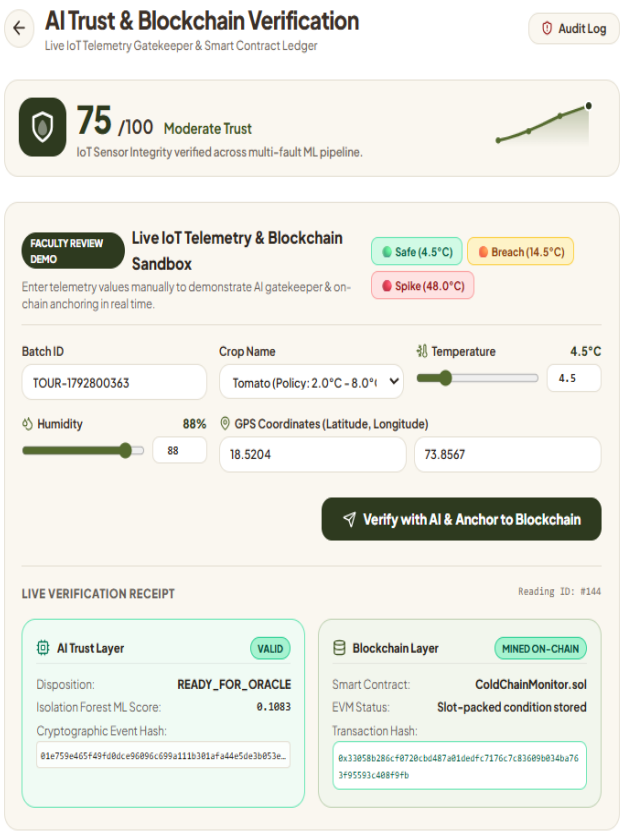


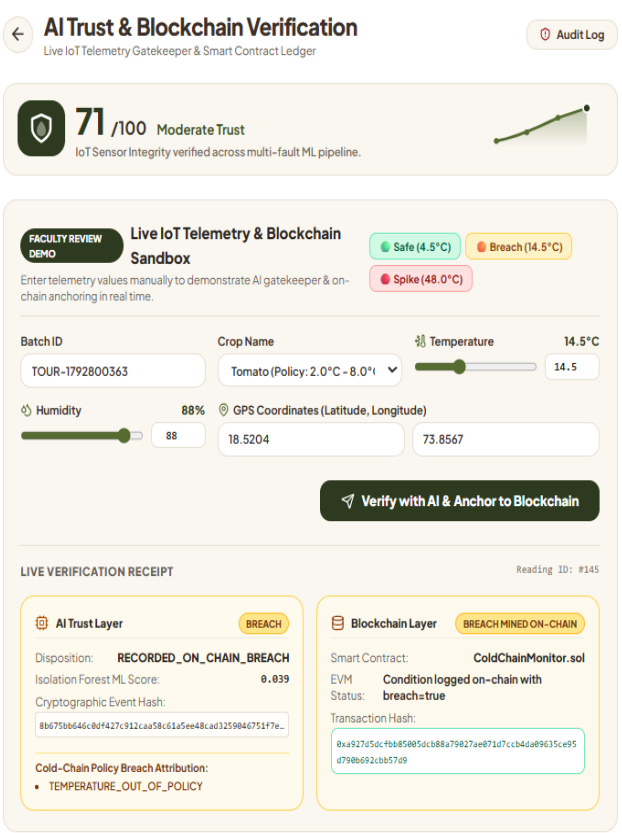
Administrative Dashboard for Supply Chain Governance and Circuit Breakers

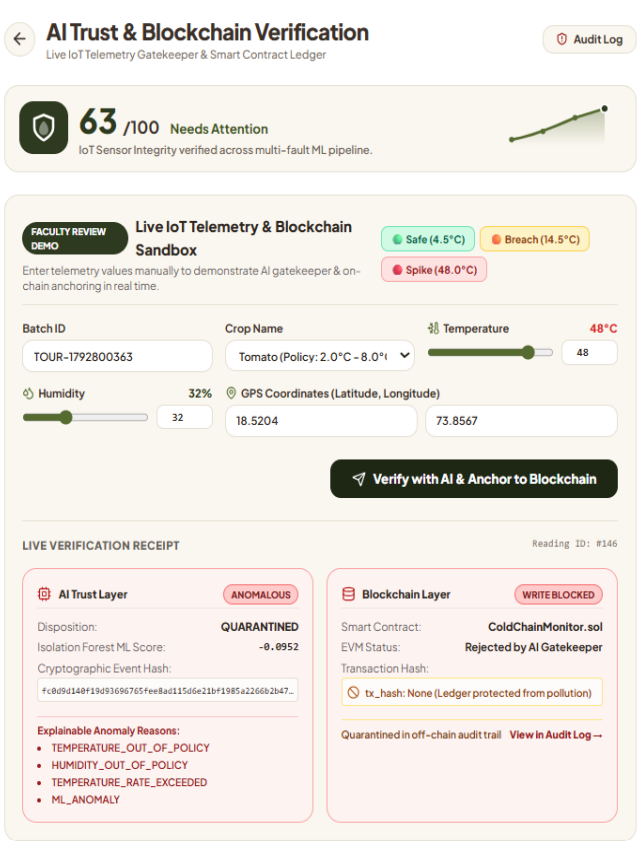


Results

AI Trust Layer for IoT Telemetry Validation and Checkpoints







Farm-to-Fork Product Journey with Cumulative Price Breakdown

Farm-to-Door Traceability Journey



Organic Produce Verified

Origin: Nashik, Maharashtra

Batch: #TOUR-1792800363

SCAN / ZOOM

This product is 100% traceable from farm to your doorstep.

View Verified Origin Certificate →

Quality Certificates & Lab Reports

AGMARK Organic Certification anchored

Inspect CIDs →

Customer Ratings & Reviews (0)

Write Review

5.0

(0 Reviews)

5★	100%
4★	0%
3★	0%
2★	0%
1★	0%

Harvested ₹1,200
20 Oct 2026 | Nashik Farm Gate

Processed & Inspected
21 Oct 2026 | Nashik Quality Hub

Transported ₹1,550
22 Oct 2026 | Safexpress Reefer Fleet

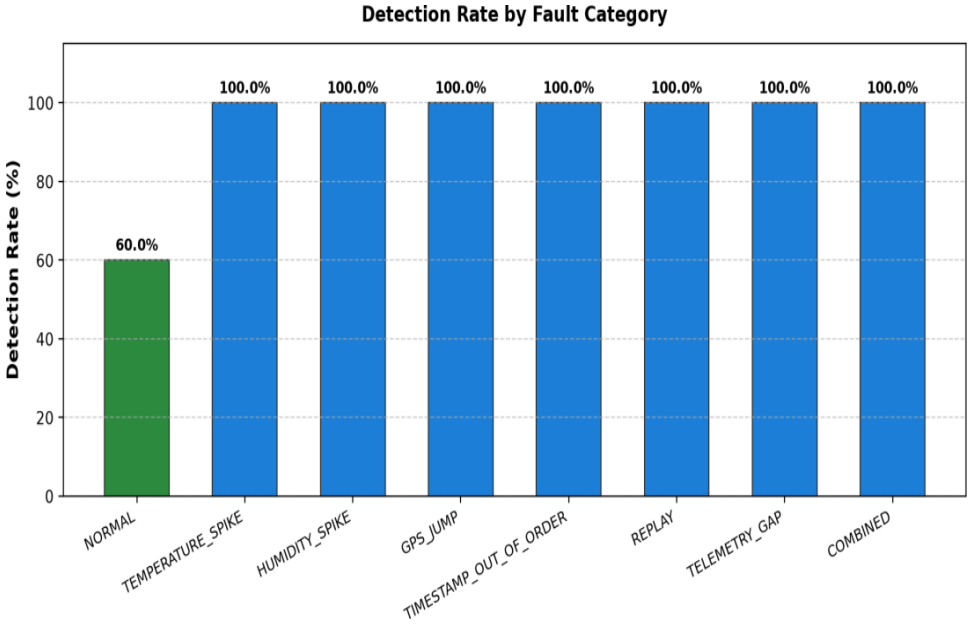
Received at Dark Store ₹1,850
23 Oct 2026 | Nashik Central Store

Out for Delivery ₹2,200
24 Oct 2026 | Last-Mile Courier

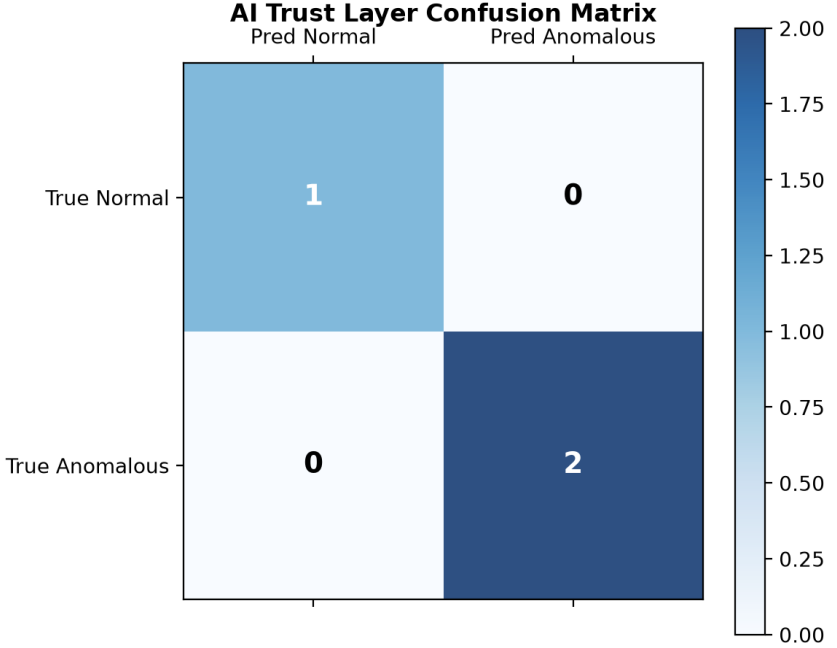
Delivered
24 Oct 2026 | Customer Doorstep

Showcase & Demo Lab

AI-Based Telemetry Anomaly Detection and Quarantine Isolation



Confusion Matrix of the AI-Based Anomaly Detection Model



Recorded End-to-End System Verification Outcomes (PRD Section 6 Suite)

Verification Area / Sub-System	Recorded Outcome	Scope of Automated Validation
Modular Smart-Contract RBAC Checks	PASS	Gating of registerBatch, transferCustody, recordCondition to assigned roles
Batched Condition Oracle Writes	PASS	EVM transaction execution of recordConditionsBatch fitting single gas slot
AI Trust Multi-Fault Evaluation	PASS	Detection of all 7 fault profiles (temp spikes, GPS teleportation, replay attacks)
Multi-Role Supply Chain Journey	PASS	State progression: REGISTERED -> IN_TRANSIT -> IN_STORAGE -> AT_RETAIL -> SOLD
IoT Hardware Breach & Push-Button Quarantine	PASS	Interception of 48 C thermal breaches and routing to quarantine hash-chain
Decentralized IPFS Quality Certificates	PASS	Pinata upload, CID generation (ipfs://Qm...), and on-chain hash anchoring
Dark Store Inbound Receiving & Checkout	PASS	GRN bay allocation, micro-inventory update, and retail checkout execution
Consumer Reviews & Rating Aggregation	PASS	Verified review submission strictly for SOLD batches with 5-star scoring
Multi-Stage Fair Price Distribution	PASS	Cumulative paise calculation verifying farmer price share transparency (50.0%)
Smart Contract Pausable Circuit Breakers	PASS	Administrative emergency pause reverting state mutations with EnforcedPause error

Future Scopes

- **Advanced AI Validation:** Deep learning for temporal trend analysis and predictive anomaly scoring.
- **Real-Time IoT Hardware:** Deploying physical edge sensors (temperature, humidity, GPS) across transit fleets.
- **Scalability & Layer-2:** Rollup architectures and off-chain storage to handle multi-regional scale at low cost.
- **Data Privacy & ZKPs:** Zero-knowledge proofs to safeguard commercial secrets while proving compliance.
- **Automated Payments:** Smart contract integration with real-world banking rails and UPI/CBDC.
- **Farmer Mobile App:** Multilingual, low-connectivity app designed for rural crop registration.
- **Govt & Market Linkages:** Connecting with e-NAM and agricultural portals for price discovery.
- **Cross-Chain Interoperability:** Interconnecting different public and enterprise blockchains.

Conclusion

- The system overcomes the blockchain-IoT Oracle dilemma by validating sensor telemetry through an **AI Trust Layer** before committing data on-chain.
- Immutable smart contracts provide verifiable, end-to-end produce traceability across strict forward-only custody stages from farm to retail.
- Automated cold-chain SLA monitoring replaces subjective manual inspections with objective, real-time penalty deductions.
- Direct smart contract escrow settlements guarantee prompt, transparent, and fair compensation for farmers without intermediary cuts.
- Consumers gain complete trust in crop origin and freshness through instant **QR-code verification** backed by real-time audit logs.

References

[1] A. Agrawal et al., "Optimizing Supply Chain Management With IoE and AI," 2025.

[2] M. Mohamed, A. Al-Saleh, A. Sharma, and M. Tejani, "Zero-Day Exploits Detection with Adaptive WavePCA-Autoencoder (AWPA) Adaptive Hybrid Exploit Detection Network (AHEDNet)," Scientific Reports, 2025.

[3] A. Maurya et al., "Blockchain-Driven Security for IoT Networks: State-of-the-Art, Challenges and Future Directions," Peer-to-Peer Networking and Applications, 2025.

[4] D. Dudczyk, M. Dunston, and M. Crosby, "Blockchain Technology for Global Supply Chain Management: A Survey of Applications, Challenges, Opportunities and Implications," IEEE Access, 2024.

[5] S. Yavaprabhas, R. Kurnia, M. Seyedghorban, and D. Samson, "Demystifying the Impact of Blockchain on Trust in Emerging and Established Relationships: A Case of Organic Food Supply Chains," 2024.

Thank You